Demand Forecasting Case Study

EXECUTIVE SUMMARY

A machine learning approach to 8-week demand forecasting for supermarket product planning.

Business Problem

- Forecast product demand 8 weeks (56 days) ahead
- Reduce stock-outs and excess inventory
- Improve production and inventory planning

Key Demand Drivers

- Historical demand — strongest predictor
- Weekly and monthly seasonality
- Promotional campaigns boosted sales
- Demand varied across supermarkets and SKUs

Forecast Performance

Best Model: Ridge Regression

- RMSE: 14.64
- R^2: 0.618
- MAPE: 5.25%

Selected for lowest RMSE with strong generalization.

Business Impact

- Improved demand planning accuracy
- Reduced stock-outs
- Lower excess inventory and write-offs
- Better production scheduling
- Improved supply chain decision-making

Conclusion: Machine learning enables accurate 8-week demand forecasting, supporting smarter inventory planning, fewer stock-outs, and lower operational costs.